

Addition of two n -bit Binary Numbers

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1 Introduction

Addition of two n -bit binary numbers is one of the fundamental arithmetic operations in digital systems. The operation is performed using n full adders connected in cascade. Before studying the complete n -bit binary adder, we first study the operation of a single full adder.

2 Full Adder

A full adder is a combinational logic circuit that adds three input bits: two operand bits, A and B , and a carry input C_{in} from the previous lower significant stage. It produces two output bits, namely the sum bit (S) and the carry bit (C_{out}).

The two output bits are necessary because the arithmetic sum of three binary digits ranges from 0 to 3, and the binary representation of 2 and 3 requires two bits.

A	B	C_{in}	S	C_{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Table 1: Truth table of a full adder.

From the truth table, the Boolean expressions for the sum and carry can be derived using Karnaugh map:

$$S = A'B'C_{\text{in}} + A'BC'_{\text{in}} + AB'C'_{\text{in}} + ABC_{\text{in}} \quad (1)$$

$$= A \oplus B \oplus C_{\text{in}}, \quad (2)$$

and

$$C_{\text{out}} = AB + AC_{\text{in}} + BC_{\text{in}}. \quad (3)$$

3 Addition of Two n -Bit Binary Numbers

Let

$$A = A_{n-1}A_{n-2} \cdots A_0$$

and

$$B = B_{n-1}B_{n-2} \cdots B_0$$

be two n -bit binary numbers.

The sum $A + B$ can be obtained by connecting n full adders sequentially, as shown in the figure below.

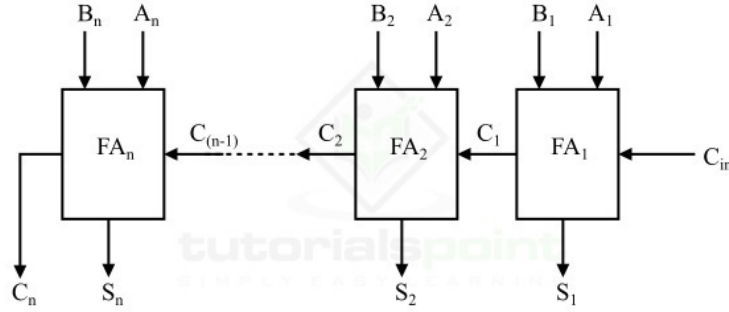


Figure 1: An n -bit binary adder using full adders.

The initial carry input is taken as

$$C_0 = 0,$$

since there is no incoming carry for the least significant bit.

The least significant bits, A_0 and B_0 , together with C_0 , are added by the first full adder. The sum output is S_0 , and the carry generated is passed as the carry input to the next full adder.

Similarly, each subsequent full adder adds the corresponding bits of the two operands together with the carry generated from the previous stage. Thus, the carry propagates from the least significant stage to the most significant stage.

The outputs of the full adders are the sum bits

$$S_0, S_1, \dots, S_{n-1},$$

and the final carry output is denoted by C_n .
If the final carry output satisfies

$$C_n = 1,$$

then an overflow has occurred. This indicates that the sum requires more than n bits for its representation.

References

1. M. Morris Mano and Michael D. Ciletti, *Digital Design*, 6th Edition, Pearson, 2018.
2. ChatGPT (OpenAI), used for LaTeX formatting, grammar editing and typesetting